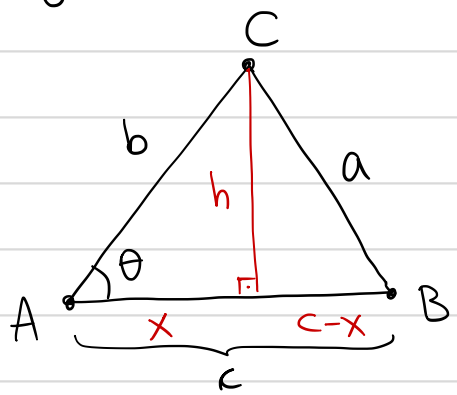
Supplementary Notes

Law of Cosines
(= Pythagorean theorem for angles other than 90°)



$$\boxed{a^2 = b^2 + c^2 - 2bc\cos\theta}$$

Proof:
$$\begin{cases} x^2 + h^2 = b^2 & (1) \\ (c-x)^2 + h^2 = a^2 & (2) \\ b\sin\theta = h & (3) \end{cases}$$

$$c^2 - 2cx + x^2 + h^2 \overset{(2)}{=} a^2$$
$$\overset{(1)}{=}$$
$$c^2 - 2c\sqrt{b^2 - h^2} + (b^2 - h^2) + h^2$$
$$\overset{(3)}{=}$$
$$c^2 - 2c\sqrt{b^2(1 - \sin^2\theta)} + b^2$$
$$=$$
$$c^2 - 2cb\cos\theta + b^2 \qquad \blacksquare$$